

_____ 13. _____ occurs when a sub-type inherits only some of the functions of a super-type and other functions are not inherited.

- A. Multiple Inheritance
- B. Single Inheritance
- C. Encapsulation
- D. Selective Inheritance
- E. None of the Above

_____ 14. Which of the following query operations is usually the most time-consuming in query optimization?

- A. Selection
- B. Projection
- C. Join
- D. Union
- E. All of the Above

_____ 15. Which of the following is not a valid step in the query optimization process?

- A. Parsing
- B. Rewriting the query
- C. Generating possible execution plans
- D. Analyzing the query history
- E. None of the Above

Part II: Say "True" if the statement is correct and False if it is incorrect.

_____ 1. One of the advantages of Object-Oriented Databases is that they support complex data types like multimedia data.

_____ 2. The query optimizer generates an execution plan only after the query has been executed

_____ 3. Object-Oriented Databases support encapsulation, meaning that an object's data and its methods are bundled together.

_____ 4. In cost-based optimization, the cost of query execution is calculated using only the number of rows involved in the query.

_____ 5. In an OODB, objects cannot be persisted; they only exist temporarily during the execution of a program.

_____ 6. In query execution, scanning a table is usually faster than using an index if the table is small.

8. _____ is moving projection operations down the query tree to eliminate unnecessary columns early.

- A. Join Reordering
- B. Selection Pushdown
- C. Combining Operations
- D. Projection Pushdown
- E. None of the Above

9. Which of the following is typically an output of query optimization?

- A. A query execution plan
- B. A simplified SQL query
- C. A data dictionary
- D. A compiled version of the query

10. _____ refers to the concept of hiding the internal workings of an object and exposing only necessary details?

- A. Polymorphism
- B. Inheritance
- C. Encapsulation
- D. Abstraction

11. Which of the following is not correct about Heuristics for Algebraic Optimization?

- A. apply first the operations that reduce the size of intermediate results.
- B. Perform select operations as early as possible to reduce the number of tuples
- C. perform project operations as early as possible to reduce the number of attribute
- D. select and join operations that are most restrictive should be executed before other similar operations
- E. None of the above

12. Which of the following is a key strategy used in query optimization?

- A. Breaking a complex query into simpler subqueries
- B. Always using a nested loop for joins
- C. Always applying a hash join for large datasets
- D. Ignoring indexes for efficiency

4. Which of the following is a primary advantage of using object-oriented databases over traditional relational databases?
- A. Faster data retrieval
 - B. Support for complex data types and relationships
 - C. Simpler schema design
 - D. Lower cost of implementation
 - E. None of the Above
5. What is the cost-based query optimization approach?
- A. An optimization strategy that tries to reduce the number of operations
 - B. An optimization strategy that evaluates the cost of different execution plans using statistical information about the data
 - C. An optimization strategy that only focuses on minimizing query syntax complexity
 - D. An optimization strategy based solely on user preferences
6. Which one of the following is not characteristics of object oriented databases?
- A. It keeps up a direct relation between real world and database objects as if objects do not lose their integrity and identity.
 - B. OODBs provide system generated object identifier for each object so that an object can easily be identified and operated upon.
 - C. OODBs are extensible, which identifies new data types and the operations to be performed on them.
 - D. Provides encapsulation, feature which, the data representation and the methods implementation are hidden from external entities.
 - E. None of the above.
7. What is the main goal of query optimization in a database management system (DBMS)?
- A. To ensure the query returns the correct results
 - B. To reduce the cost of executing a query
 - C. To convert SQL queries into machine language
 - D. To create a backup of the database
 - E. None of the Above

15%)	II(6%)	III(9%)	Total (30%)

Addis Ababa University
Department of Software Engineering
Advanced Database Systems
Mid Examination (30%)

Name _____

Time Allotted: 1:10 Hr.
Date: December, 2024

ID _____

Part I: Choose the best answer from the given alternative. (Each 1 Mark)

1. Among the following which one is not the weakness of relational database?

- A. uses a single data type
- B. No Direct Support for Hierarchies
- C. Ensure data from existing applications is stored and managed persistently
- D. Not suitable for implementing real world problems
- E. None of the Above

2. Which of the following is NOT typically considered during the query optimization process?

- A. Evaluating different join algorithms
- B. Choosing an appropriate index for a table
- C. Rewriting the query to a simpler form
- D. Ensuring the query is free of syntax errors
- E. None of the Above

3.

Suppose want to list all loans and amount of loans, but do not care about the branch name which one of the following allows us to produce this relation

- A. Selection
- B. set difference
- C. Union
- D. Cartesian product
- E. Projection

I(15%)	II(6%)	III(9%)	Total (30%)

Addis Ababa University
Department of Software Engineering
Advanced Database Systems
Mid Examination (30%)

Name _____

Time Allotted: 1:10 Hr.
Date: December, 2024

ID _____

Part I: Choose the best answer from the given alternative. (Each 1 Mark)

1. Among the following which one is not the weakness of relational database?

- A. uses a single data type
- B. No Direct Support for Hierarchies
- C. Ensure data from existing applications is stored and managed persistently
- D. Not suitable for implementing real world problems
- E. None of the Above

2. Which of the following is NOT typically considered during the query optimization process?

- A. Evaluating different join algorithms
- B. Choosing an appropriate index for a table
- C. Rewriting the query to a simpler form
- D. Ensuring the query is free of syntax errors
- E. None of the Above

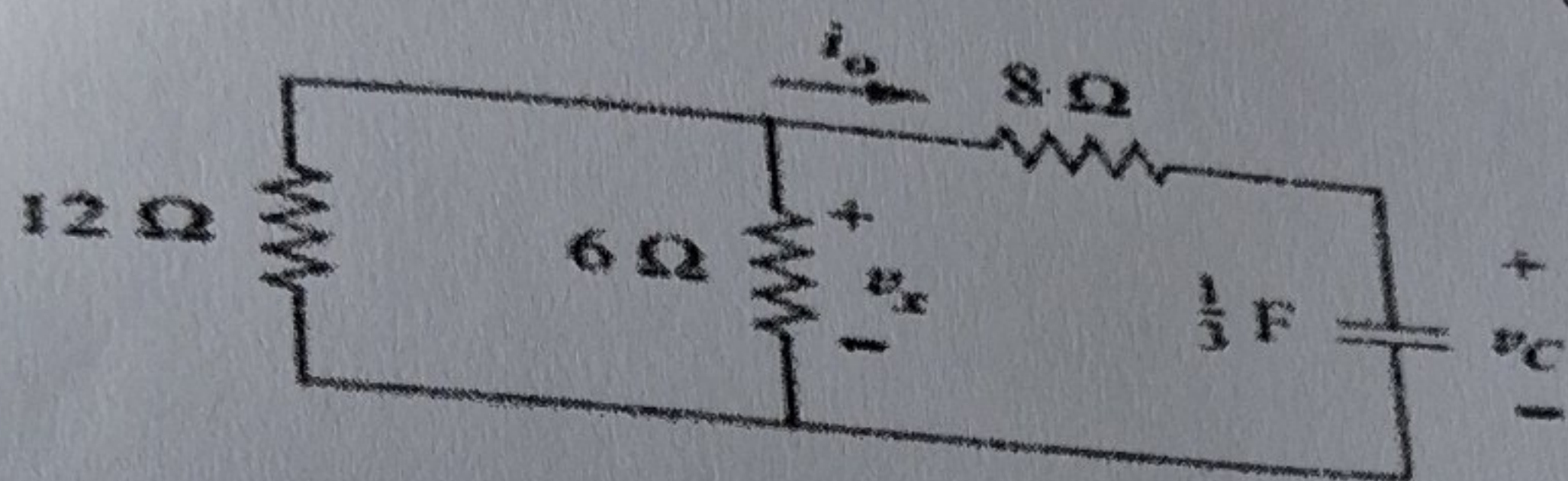
3.

Suppose want to list all loans and amount of loans, but do not care about the branch name which one of the following allows us to produce this relation

- A. Selection
- B. set difference
- C. Union

- D. Cartesian product
- E. Projection

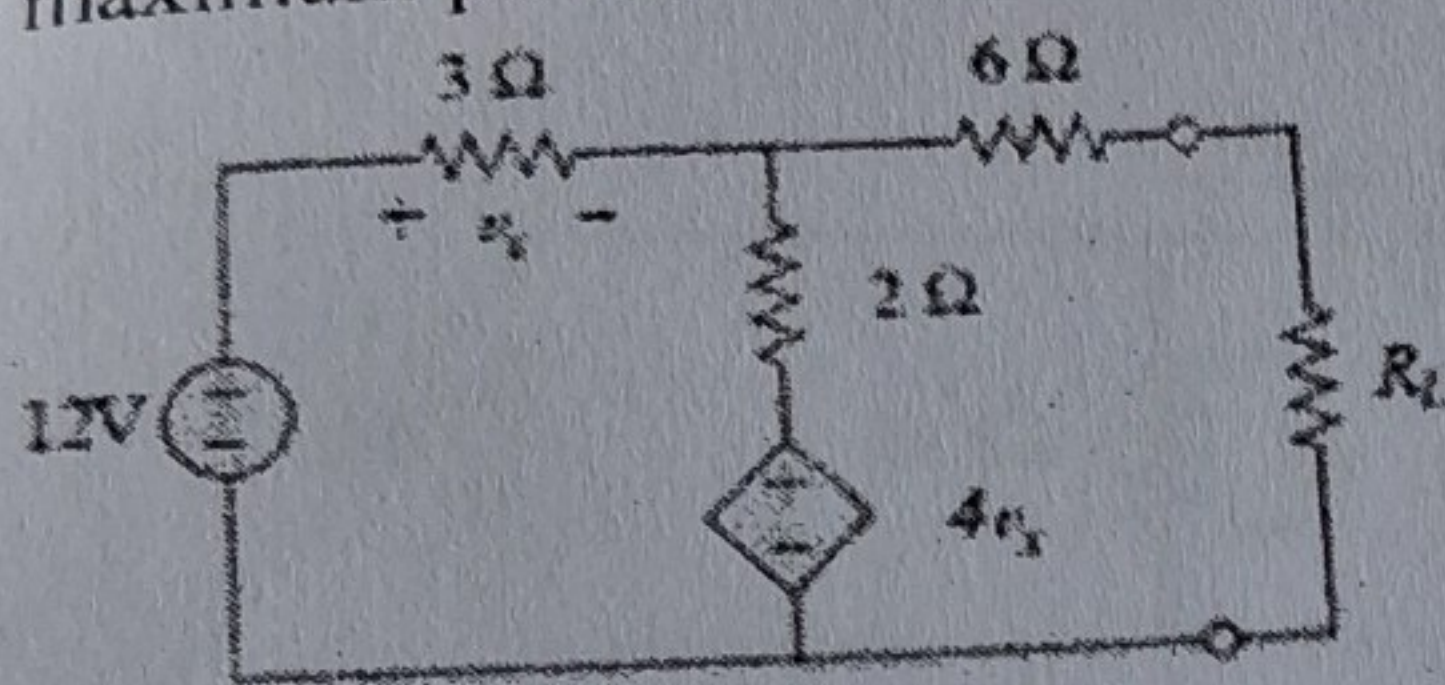
3. Determine V_C , V_x and i_0 for the circuit shown below if $V_C(0) = 45V$



[4pts]

[6pts]

Determine the value of that will draw the maximum power from the rest of the circuit in figure below. Calculate the maximum power [6pts]





Name: _____

ID No: _____

- Write your answer Briefly
- Try all questions
- Write your name on

May 10, 2026

Time allowed 2:00hrs

Test #2. (Marks 25%)

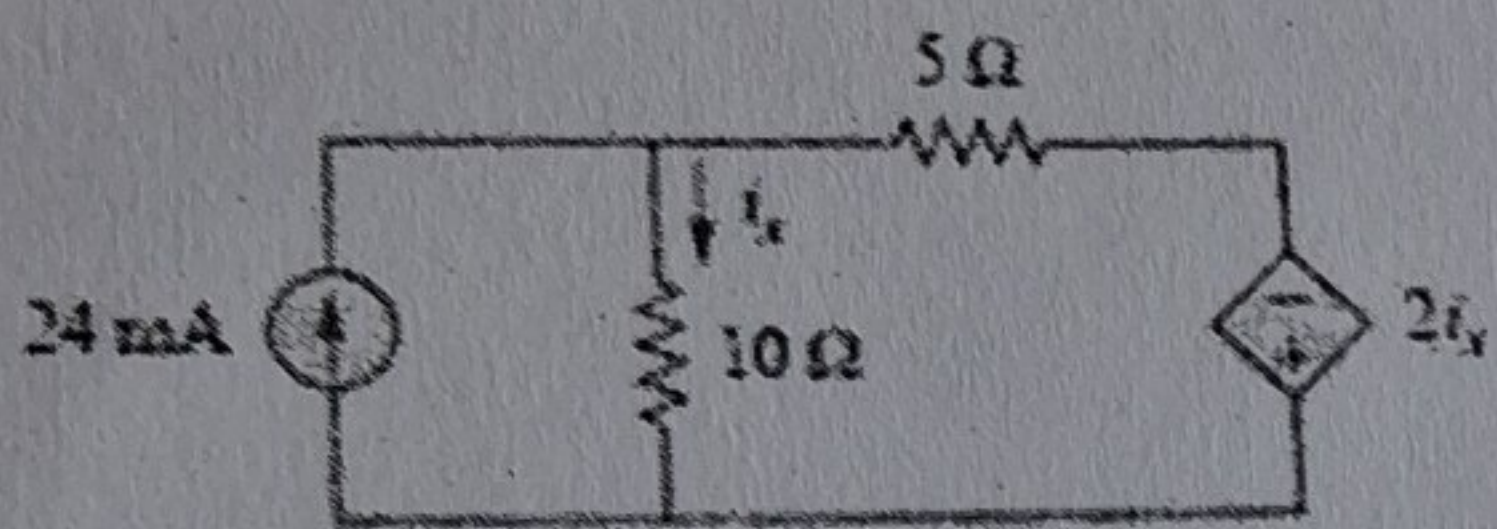
PART I Multiple Choice (0.5pts Each) Total [5Pts]

1. The Superposition Principle is specifically applicable to
 - A) All circuits regardless of components
 - B) Linear circuits only
 - C) Circuits with only one independent source
 - D) High-power AC circuits only
2. When "turning off" an independent current source for analysis, you should replace it with
 - A) A short circuit
 - B) An open circuit
 - C) A 1Ω resistor
 - D) A wire
3. A source transformation allows you to convert a current source in parallel with a resistor into
 - A) A current source in series with a resistor
 - B) A voltage source in parallel with a resistor
 - C) A voltage source in series with a resistor
 - D) A dependent source
4. The Thévenin Resistance R_{Th} is found by
 - A) Dividing open-circuit voltage by short-circuit current
 - B) Multiplying voltage by current
 - C) Summing all resistances in the circuit blindly
 - D) Measuring the voltage across a load
5. To achieve Maximum Power Transfer to a load (R_L), the load resistance must be
 - A) Zero
 - B) Twice the internal resistance
 - C) Equal to the Thévenin equivalent resistance
 - D) As large as possible
6. When inductors are connected in series, they behave mathematically like
 - A) Capacitors in series
 - B) Resistors in series
 - C) Resistors in parallel
 - D) Diodes in series
7. An inductor opposes sudden changes in
 - A) Voltage
 - B) Resistance
 - C) Current
 - D) Temperature
8. In a DC steady-state circuit (after a long time), a capacitor acts as
 - A) A short circuit
 - B) An open circuit
 - C) A battery
 - D) A resistor
9. In a source-free RL circuit, the current decays
 - A) Linearly
 - B) Sinusoidally
 - C) Exponentially
 - D) It does not decay
10. A "Step Response" typically occurs when
 - A) A switch is moved to connect a DC source
 - B) The circuit is turned off
 - C) The resistor is removed
 - D) The circuit reaches 5 time constants

PART II ANALYTICAL QUESTIONS [20pts]

1. Use source transformation to find i_x in the circuit shown below

[4pts]



- Full Name
- Age
- Gender (Radio Buttons or Spinner)
- Contact Number
- Address
- Illness Description
- Registration Date (auto-filled or date picker)

- Store registered patient data using SQLite database.
- Success & Error Feedback- Toast messages after actions

The following image shows a sample interface for the registration form.

Part V: Code Writing

1. Design and implement a **Currency Converter** mobile application for Android. The app should allow users to:

- Input an amount in a chosen source currency.
- Select both the source currency and the target currency from dropdown lists (Spinners).
- Convert the entered amount into the target currency based on a set of predefined exchange rates.
- Display the converted amount clearly on the screen.
- Handle invalid inputs gracefully (e.g., empty or non-numeric values).
- *(Optional)* Include a button to swap the source and target currencies.

Additional Requirements:

- Use appropriate Android UI components (EditText, Spinner, Button, TextView).
- Implement the logic for currency conversion inside the app (you may hardcode exchange rates or use a simple local data source).
- Ensure the UI is user-friendly and responsive.

Contact

Currency Converter

From

Algeria

To

Brunei

Amount

020

Convert

Result

BRL

Converted Amount

2. Develop an android application for **Patient Registration System** having the following Features:

1. Create the Patient Registration form with fields:

Part III: Fill in the blank space provided below. (Each 1 Mark)

1. In order to get the user's current location, we use the _____ class.
2. Android UI elements like Button, TextView, and EditText are subclasses of the _____ class.
3. _____ layout allows you to define positions of child views relative to each other.
4. The class _____ is used to manage database creation and version management.
5. To display a map in an Android app, you usually use the _____ widget in XML.

Part IV: Write the brief description for the answer of each question as required.

1. Compare and contrast LinearLayout, RelativeLayout, and ConstraintLayout. Under what circumstances would you choose one over the others? (3 pts.)
2. Describe the purpose and structure of the **SQLiteOpenHelper** class. Include in your explanation how to create a database, insert data, and perform queries. (2 pts.)

Part III: Fill in the blank space provided below. (Each 1 Mark)

1. In order to get the user's current location, we use the _____ class.
2. Android UI elements like Button, TextView, and EditText are subclasses of the _____ class.
3. _____ layout allows you to define positions of child views relative to each other.
4. The class _____ is used to manage database creation and version management.
5. To display a map in an Android app, you usually use the _____ widget in XML.

Part IV: Write the brief description for the answer of each question as required.

1. Compare and contrast LinearLayout, RelativeLayout, and ConstraintLayout. Under what circumstances would you choose one over the others? (3 pts.)
2. Describe the purpose and structure of the **SQLiteOpenHelper** class. Include in your explanation how to create a database, insert data, and perform queries. (2 pts.)

6. Among the following which one correctly describes View Group in Android?
 - A. A special TextView subclass
 - B. A layout that cannot contain any other views
 - C. A container that holds other views
 - D. A widget for displaying images
 - E. None of the Above
7. Which Google API is used for maps in Android applications?
 - A. GoogleMapApi
 - B. MapsAPIv3
 - C. Google Maps SDK for Android
 - D. AndroidGeoAPI
8. Which of the following is a subclass of ViewGroup?
 - A. TextView
 - B. ImageView
 - C. LinearLayout
 - D. Button
9. What class is used to read data row-by-row from an SQLite query?
 - A. DataReader
 - B. Cursor
 - C. Scanner
 - D. Iterator
10. Which layout arranges its children in a single column or row?
 - A. FrameLayout
 - B. RelativeLayout
 - C. GridLayout
 - D. LinearLayout
 - E. All of the Above
11. Which object is used to store a set of values for database insertion?
 - A. HashMap
 - B. Intent
 - C. Bundle
 - D. ContentValues
12. What is required in the Android manifest to use location services?
 - A. `<uses-permission android:name="android.permission.LOCATION" />`
 - B. `<uses-permission android:name="android.permission.GET_LOCATION" />`
 - C. `<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />`
 - D. `<uses-permission android:name="android.permission.MAP_SERVICE" />`
13. What is the base class for all UI components in Android?
 - A. ViewGroup
 - B. View
 - C. Activity
 - D. Layout
14. Which method is called only once when the database is created for the first time?
 - A. onCreate()
 - B. onStart()
 - C. onUpgrade()
 - D. onInsert()
15. What is the correct way to set a click listener on a button in XML?
 - A. `<Button android:click="myMethod" />`
 - B. `<Button android:onClick="myMethod" />`
 - C. `<Button setOnClickListener="myMethod" />`
 - D. `<Button onClickListener="myMethod" />`
16. Which layout is best for performance when dealing with deeply nested UI elements?
 - A. LinearLayout
 - B. FrameLayout
 - C. ConstraintLayout
 - D. TableLayout
17. _____ is a layer contains the core C++ libraries that make up the primitives that support Flutter apps.
 - A. Framework
 - B. Embedder
 - C. Engine
 - D. Plugin

6. Among the following which one correctly describes View Group in Android?
 - A. A special TextView subclass
 - B. A layout that cannot contain any other views
 - C. A container that holds other views
 - D. A widget for displaying images
 - E. None of the Above
7. Which Google API is used for maps in Android applications?
 - A. GoogleMapApi
 - B. MapsAPIv3
 - C. Google Maps SDK for Android
 - D. AndroidGeoAPI
8. Which of the following is a subclass of ViewGroup?
 - A. TextView
 - B. ImageView
 - C. LinearLayout
 - D. Button
9. What class is used to read data row-by-row from an SQLite query?
 - A. DataReader
 - B. Cursor
 - C. Scanner
 - D. Iterator
10. Which layout arranges its children in a single column or row?
 - A. FrameLayout
 - B. RelativeLayout
 - C. GridLayout
 - D. LinearLayout
 - E. All of the Above
11. Which object is used to store a set of values for database insertion?
 - A. HashMap
 - B. Intent
 - C. Bundle
 - D. ContentValues
12. What is required in the Android manifest to use location services?
 - A. `<uses-permission android:name="android.permission.LOCATION" />`
 - B. `<uses-permission android:name="android.permission.GET_LOCATION" />`
 - C. `<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />`
 - D. `<uses-permission android:name="android.permission.MAP_SERVICE" />`
13. What is the base class for all UI components in Android?
 - A. ViewGroup
 - B. View
 - C. Activity
 - D. Layout
14. Which method is called only once when the database is created for the first time?
 - A. onCreate()
 - B. onStart()
 - C. onUpgrade()
 - D. onInsert()
15. What is the correct way to set a click listener on a button in XML?
 - A. `<Button android:click="myMethod" />`
 - B. `<Button android:onClick="myMethod" />`
 - C. `<Button setOnClickListener="myMethod" />`
 - D. `<Button onClickListener="myMethod" />`
16. Which layout is best for performance when dealing with deeply nested UI elements?
 - A. LinearLayout
 - B. FrameLayout
 - C. ConstraintLayout
 - D. TableLayout
17. _____ is a layer contains the core C++ libraries that make up the primitives that support Flutter apps.
 - A. Framework
 - B. Embedder
 - C. Engine
 - D. Plugin

Part V: Code Writing

1. Design and implement a **Currency Converter** mobile application for Android. The app should allow users to:

- Input an amount in a chosen source currency
- Select both the source currency and the target currency from dropdown lists (Spinners)
- Convert the entered amount into the target currency based on a set of predefined exchange rates.
- Display the converted amount clearly on the screen
- Handle invalid inputs gracefully (e.g., empty or non-numeric values).
- (Optional) Include a button to swap the source and target currencies.

Additional Requirements:

- Use appropriate Android UI components (EditText, Spinner, Button, TextView).
- Implement the logic for currency conversion inside the app (you may hardcode exchange rates or use a simple local data source).
- Ensure the UI is user-friendly and responsive.

CONTACT

Currency Converter

From
USD

To
EUR

Amount
100

Convert

Result
Converted Amount
95

2. Develop an android application for Patient Registration System having the following Features:
1. Create the Patient Registration form with fields:

18. Which class is primarily used to access location services in Android?
- A. LocationProvider
 - B. LocationManager
 - C. LocationTracker
 - D. MapManager
19. What does wrap_content do in a layout parameter?
- A. Hides the view
 - B. Fills the entire screen
 - C. Sets the size to zero
 - D. Sizes the view just big enough to fit its content
 - E. None of the Above
20. Which callback method is triggered when the map is ready to be used?
- A. onMapInitialize()
 - B. onMapReady()
 - C. initMap()
 - D. loadMap()
21. Which class is used to represent a writable SQLite database?
- A. SQLiteDatabase
 - B. SQLiteHelper
 - C. Cursor
 - D. ContentValues
22. The purpose of setState() method in Flutter is to _____
- A. Exits the app
 - B. Navigates to another screen
 - C. Rebuilds the widget tree
 - D. Deletes data
23. Which method is used to add a marker on a Google Map?
- A. addMarker()
 - B. placeMarker()
 - C. setMarker()
 - D. createMarker()
24. What method is typically overridden to handle touch events manually in a custom View?
- A. dispatchTouchEvent()
 - B. onClick()
 - C. onTouch()
 - D. onTouchEvent()
 - E. None of the Above
25. Which widget displays a scrollable list?
- A. Column
 - B. GridView
 - C. ListView
 - D. Stack

Part II: Say True if the statement is correct otherwise say False if it is incorrect. (Each 1 Mark)

- _____ 1. RecyclerView contains a list of elements and enables scrolling by adding and removing elements dynamically.
- _____ 2. The LocationListener interface updates location only when the app is in the foreground.
- _____ 3. Android requires only the INTERNET permission to display Google Maps.
- _____ 4. Widgets in Flutter are immutable and cannot be modified after creation.
- _____ 5. Flutter's "Hot Reload" feature allows developers to see UI changes through restarting the app.

18. Which class is primarily used to access location services in Android?
- | | |
|---------------------|--------------------|
| A. LocationProvider | C. LocationTracker |
| B. LocationManager | D. MapManager |
19. What does wrap_content do in a layout parameter?
- Hides the view
 - Fills the entire screen
 - Sets the size to zero
 - Sizes the view just big enough to fit its content
 - None of the Above
20. Which callback method is triggered when the map is ready to be used?
- | | |
|----------------------|--------------|
| A. onMapInitialize() | C. initMap() |
| B. onMapReady() | D. loadMap() |
21. Which class is used to represent a writable SQLite database?
- | | |
|-------------------|------------------|
| A. SQLiteDatabase | C. Cursor |
| B. SQLiteHelper | D. ContentValues |
22. The purpose of setState() method in Flutter is to _____
- Exits the app
 - Navigates to another screen
 - Rebuilds the widget tree
 - Deletes data
23. Which method is used to add a marker on a Google Map?
- | | |
|------------------|-------------------|
| A. addMarker() | C. setMarker() |
| B. placeMarker() | D. createMarker() |
24. What method is typically overridden to handle touch events manually in a custom View?
- | | |
|-------------------------|----------------------|
| A. dispatchTouchEvent() | D. onTouchEvent() |
| B. onClick() | E. None of the Above |
| C. onTouch() | |
25. Which widget displays a scrollable list?
- | | |
|-------------|-------------|
| A. Column | C. ListView |
| B. GridView | D. Stack |

Part II: Say True if the statement is correct otherwise say False if it is incorrect. (Each 1 Mark)

- ____ 1. RecyclerView contains a list of elements and enables scrolling by adding and removing elements dynamically.
- ____ 2. The LocationListener interface updates location only when the app is in the foreground.
- ____ 3. Android requires only the INTERNET permission to display Google Maps.
- ____ 4. Widgets in Flutter are immutable and cannot be modified after creation.
- ____ 5. Flutter's "Hot Reload" feature allows developers to see UI changes through restarting the app.

Addis Ababa University
Department of Software Engineering
Mobile Application Development
Final Examination (50%)

Name: _____

ID: _____

Date: May 27, 2025

Time Allotted: 120 Minutes

Instructions:

1. Switch off your mobile phone.
2. Read the questions carefully and attempt all.
3. Make sure that the exam has 6 pages.
4. Answers that are **not legibly** written will not be marked.
5. Any form of cheating will nullify the result of both parties

Part I: Choose the best answer from the given alternative. (Each 1 Mark)

1. What does *android:layout_weight* do in a *LinearLayout*?
 - A. Sets the importance of a view
 - B. Defines the order of views
 - C. Makes a view invisible
 - D. Distributes extra space proportionally among views
 - E. All of the Above
2. What attribute is used to set the width of a view to match its parent?
 - A. *android:width="fill_parent"*
 - B. *android:layout_width="match_parent"*
 - C. *android:width="wrap_content"*
 - D. *android:width="0dp"*
3. Which of the following best describes a *StatelessWidget*?
 - A. A widget that holds mutable state
 - B. A widget that rebuilds when state changes
 - C. A widget with immutable properties
 - D. A widget used for navigation
4. Which attribute sets the width of a view to fill the parent container?
 - A. *android:width="wrap"*
 - B. *android:layout_width="match_parent"*
 - C. *android:layout_width="wrap_content"*
 - D. *android:layout_height="fill"*
5. What does the hot reload feature in Flutter do?
 - A. Restarts the app from scratch
 - B. Compiles Dart to native code
 - C. Clears the app cache
 - D. Updates code and reflects changes without restarting the app

- A. Confirmation bias: Rely on logical necessity rather than real-world testing failure
- B. Deductive hypothesis: Rely on logical necessity rather than critical testing
- C. Paradigm-based theories: Reflect consensus rather than critical testing
- D. Unfalsifiable hypotheses: Cannot be tested or potentially proven wrong through observation
- E. Circular hypotheses: Use their own conclusions as their justification

10. An astronomer proposes the hypothesis: "If black holes collide, they will produce gravitational waves detectable by instruments like LIGO." This hypothesis is scientific because it makes a clear, testable prediction. If no gravitational waves are detected after a black hole merger is observed, the hypothesis would be proven false. Its strength lies not in repeated confirmation, but in the fact that it can be challenged by observable evidence. Question: Why does this black hole hypothesis meet the criteria for a scientific theory under Karl Popper's view?

- A. Because it fits well with Einstein's theory of general relativity and has been confirmed by physicists.
 - B. Because it describes a complex phenomenon that cannot be directly tested but is widely accepted.
 - C. Because it makes a prediction that can be tested by experiment and could be shown to be false by observational data.
 - D. Because it has been proven correct through numerous successful detections of gravitational waves.
11. Which of the following best reflects Karl Popper's critique of inductive reasoning and his alternative approach to scientific progress?
- A. Scientific theories should be built from repeated observations and generalized into universal laws through inductive logic.
 - B. Because we cannot observe all instances in the universe, generalizing from a limited set of observations is unreliable; instead, science should focus on making testable predictions that can be refuted.
 - C. Inductive reasoning is more rigorous than deductive reasoning because it relies on actual empirical data.
 - D. A theory is scientific only if it has been verified many times through repeated confirmation in experiments.

12. In Karl Popper's view, when a scientific hypothesis is falsified, it is not considered a failure. Instead, such falsification is valuable because it:
- A. Proves that science should rely more on confirmation than criticism
 - B. Confirms the superiority of well-established scientific paradigms
 - C. Validates the use of induction as the foundation of science
 - D. Highlights the limitations of existing theories and opens new directions for inquiry
13. Who is credited with advertising or popularizing the term "induction" in the development of scientific methodology?
- A. Karl Popper
 - B. Francis Bacon
 - C. David Hume
 - D. Francis Bertrand
14. Which one of the following are the two proposed solutions to David Hume's problem of induction?
- A. Inductive reasoning is strictly logical, and its validity is guaranteed by formal proof; alternatively, it is justified by the metaphysical certainty of natural laws.
 - B. Inductive reasoning is a psychological habit formed by past success; alternatively, it is pragmatically justified because it effectively predicts future events.
 - C. Inductive reasoning is infallible due to empirical evidence; alternatively, it is invalid and should be replaced by deductive methods.
 - D. Inductive reasoning is rejected entirely; alternatively, it is considered a tautology in scientific inquiry.
15. Karl Popper's concept of falsifiability is central to distinguishing scientific theories from non-scientific ones. Which of the following best captures the essence of this criterion?
- A. A scientific theory must be formulated so that it can never be disproven, ensuring its permanence and universal applicability in all future observations.
 - B. A scientific hypothesis should be structured to allow for empirical tests that could, at least in principle, demonstrate its falsehood, thereby enabling critical scrutiny and potential rejection based on evidence.
 - C. The hallmark of scientific theories is that they are supported by inductive evidence accumulated over time, making them increasingly probable as more data is gathered.
 - D. Scientific propositions should be accepted when they align with metaphysical beliefs and provide explanations that transcend empirical testing.
 - E. A hypothesis qualifies as scientific only if it has been consistently verified through repeated observations and experiments, affirming its absolute correctness.

- B. A physicist uses Einstein's theory of relativity to calculate the time dilation experienced by astronauts in orbit.
- C. An engineer tests a hypothesis about structural integrity by simulating extreme weather conditions in a lab.
- D. A biologist accepts a theory because it has been published in a peer-reviewed journal.
7. An astronomer looks through a telescope and sees a bright streak across the night sky. This vivid sensory experience of the meteor is a _____. Later, when recalling that sight from memory, the image in the astronomer's mind is a _____, a less vivid copy of the original experience. Question: Which pair of terms correctly fills the blanks according to David Hume?
- A. Perception; Sensation
- B. Idea; Impression
- C. Sensation; Perception
- D. Impression; Idea
8. A chemist has consistently observed that when two specific substances are mixed under controlled conditions, a reaction occurs producing heat. Based on these repeated observations, the chemist expects the same reaction to happen in future experiments. However, Hume questions whether this expectation is rationally justified. Why does David Hume argue that the chemist's expectation that mixing the substances will always produce heat is not logically certain?
- A. Because scientific predictions are always true if based on repeated experiments.
- B. Because chemical reactions are always completely unpredictable and random.
- C. Because the chemist's prediction is based solely on past experiences without a rational guarantee that nature's patterns will continue unchanged.
- D. Because the chemist understands the absolute laws governing heat production in reactions.
9. Imagine a scientist proposes a theory that a mysterious cosmic force subtly controls all future events in the universe but the force leaves no observable traces, cannot be measured, and appropriately explains any outcome, no matter how unexpected. Despite skepticism, no test is ever able to refute the theory because every result is interpreted as confirmation of it. This situation exemplifies Popper's belief that _____ hypotheses are problematic because they;

3. Which of the following best captures the three central questions of science?

- A. Understanding how scientific knowledge progresses through evidence and how scientific methods ensure objectivity
- B. Investigating how scientific theories reflect reality and whether they can ever offer absolute truth
- C. Examining how science interacts with society, including its role in shaping cultural values and ethical decisions.
- D. Exploring what counts as science, how reliable scientific knowledge is, and what science ultimately aims to achieve.

4. Karl Popper's philosophy of science emphasizes the importance of falsifiability in scientific theories. Which of the following best reflects Popper's response to the problem of induction?

- A. Popper argued that induction is a reliable method for establishing scientific truths, as it allows for the generalization of observed patterns to unobserved cases.
- B. Popper rejected the problem of induction, asserting that scientific theories can be conclusively proven through repeated empirical observations.
- C. Popper acknowledged the problem of induction and proposed that scientific theories can only be provisionally corroborated through rigorous testing and attempted falsification.
- D. Popper viewed deduction and induction as unnecessary for scientific inquiry, as he believed that scientific theories should be derived solely from deductive reasoning and logical analysis.

5. Demarcation Criterion of scientific claims according to Karl Popper are

- | | | |
|-------------------|----------------------|-------------------|
| A. Falsifiability | C. Verification | E. All Except 'D' |
| B. Testability | D. Empirical Content | |

6. Which of the following scenarios best demonstrates inductive reasoning in science?

- A. After observing that patients who take a new drug consistently show improved symptoms, a medical researcher concludes that the drug is likely effective for treating the condition.

3. Which of the following best captures the three central questions of the philosophy of science?

- A. Understanding how scientific knowledge progresses through evidence and how scientific methods ensure objectivity
- B. Investigating how scientific theories reflect reality and whether they can ever offer absolute truth
- C. Examining how science interacts with society, including its role in shaping cultural values and ethical decisions
- D. Exploring what counts as science, how reliable scientific knowledge is, and what science ultimately aims to achieve

4. Karl Popper's philosophy of science emphasizes the importance of falsifiability in scientific theories. Which of the following best reflects Popper's response to the problem of induction?

- A. Popper argued that induction is a reliable method for establishing scientific truths, as it allows for the generalization of observed patterns to unobserved cases.
- B. Popper rejected the problem of induction, asserting that scientific theories can be conclusively proven through repeated empirical observations.
- C. Popper acknowledged the problem of induction and proposed that scientific theories can only be provisionally corroborated through rigorous testing and attempted falsification.
- D. Popper viewed deduction and induction as unnecessary for scientific inquiry, as he believed that scientific theories should be derived solely from deductive reasoning and logical analysis.

5. Demarcation Criterion of scientific claims according to Karl Popper are

- A. Falsifiability
- B. Testability
- C. Verification
- D. Empirical Content
- E. All Except 'D'

6. Which of the following scenarios best demonstrates inductive reasoning in science?

- A. After observing that patients who take a new drug consistently show improved symptoms, a medical researcher concludes that the drug is likely effective for treating the condition.

PART I: Multiple Choices: Choose the Correct Answer and Write the Letter of Your Choice in Capital Letter. (1.5 pt. each).

1. Which of the following statements accurately explains the features and aims of the philosophy of science?
 - A. Philosophy of science investigates the methods and principles employed in scientific inquiry, aiming to understand the nature of scientific knowledge and its limits.
 - B. Philosophy of science focuses on exploring the historical development of scientific theories and their social impacts, without addressing their epistemological foundations.
 - C. Philosophy of science primarily concerns itself with promoting scientific consensus and validating established scientific facts, and uncritically evaluating scientific methodologies.
 - D. Philosophy of science is primarily concerned with metaphysical speculation and abstract philosophical inquiries, rather than engaging with empirical observations and scientific experiments.
2. All of the following accurately explain the theory of verification principle of the logical empiricists (logical positivism), except?
 - A. A statement is meaningful and scientific if and only if it can be empirically verified.
 - B. Analytic statements are true by definition and their truth can be determined solely through an analysis of the meanings of the terms involved, without reference to empirical evidence.
 - C. Ethical and religious statements are neither verifiable nor falsifiable through empirical evidence, and therefore, they are deemed as lacking meaningful content from a scientific standpoint.
 - D. Scientific statements are purely subjective and lack objective validity, rendering them unreliable for making meaningful claims about the natural world.
 - E. C and D